XRPL Validator — Daily Engineering Report · 2026-09-05

Period: 2026-09-05 00:00 to 24:00 US Eastern (2026-09-05 04:00 to 2026-09-06 04:00 UTC); ledgers #106,770,382 to #106,792,707. Every count from the ledger and the validator's logs stops at #106,792,707. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 38 engine receipts recorded (18 fixed, 0 explained, 20 without a recorded fix).
- **Changes:** 0 deployments, 25 commits merged into main, 4 engine findings fixed (F181–F184).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,770,382–#106,792,707), by step:

Step	Ledgers with a receipt
Step 1 · native engine	38
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,770,629	Step 1 · native engine	25 ledger change(s) the network made were absent from ours: CFEC8953 (deleted), D5533A8D (deleted), CFEC8953 (deleted); 27 object(s) differed in content	fixed: commit dda33ab, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-02	—	#106,771,926	Step 1 · native engine	5 object(s) differed in content	no disposition recorded
R-03	—	#106,771,950	Step 1 · native engine	6 ledger change(s) the network made were absent from ours: 0339209F, 48B655A7, 5A792787; 2 object(s) differed in content	fixed: commit 20bd643, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-04	—	#106,771,970	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-05	—	#106,772,087	Step 1 · native engine	6 ledger change(s) the network made were absent from ours: 5A792787, 8250CE37, 48B655A7; 2 object(s) differed in content	fixed: commit 20bd643, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-06	—	#106,772,093	Step 1 · native engine	6 ledger change(s) the network made were absent from ours: 337A6627 (deleted), 48B655A7, DFA3B6DD (deleted); 2 object(s) differed in content	fixed: commit 20bd643, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-07	—	#106,772,946	Step 1 · native engine	1 object(s) differed in content	fixed: commit dd4c94e, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-08	—	#106,773,044	Step 1 · native engine	1 object(s) differed in content	fixed: commit 48928e9, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-09	—	#106,773,978	Step 1 · native engine	1 object(s) differed in content	fixed: commit dd4c94e, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-10	—	#106,774,713	Step 1 · native engine	3 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: commit 51a6221, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-11	—	#106,774,868	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 5059AAF9, 0DA332A5, 88072A29; 3 change(s) of ours the network did not make; 2 object(s) differed in content	no disposition recorded
R-12	—	#106,777,783	Step 1 · native engine	2 object(s) differed in content	fixed: commit dd4c94e, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-13	—	#106,778,558	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-14	—	#106,779,252	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	fixed: commit 039e57e, deployed in cycle 121 (2026-09-11 17:39 UTC)

ID	Time	Ledger(s)	Step	Description	Disposition
R-15	—	#106,779,295	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-16	—	#106,779,421	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-17	—	#106,779,797	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make	no disposition recorded
R-18	—	#106,779,825	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-19	—	#106,779,836	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-20	—	#106,779,890	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make	no disposition recorded
R-21	—	#106,780,372	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make	no disposition recorded
R-22	—	#106,780,685	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-23	—	#106,780,690	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-24	—	#106,781,871	Step 1 · native engine	1 object(s) differed in content	fixed: commit ceabbf8, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-25	—	#106,782,285	Step 1 · native engine	2 change(s) of ours the network did not make; 4 object(s) differed in content	fixed: commit 188fda9, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-26	—	#106,782,303	Step 1 · native engine	1 object(s) differed in content	fixed: commit ceabbf8, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-27	—	#106,782,599	Step 1 · native engine	8 ledger change(s) the network made were absent from ours: 0F772B99, 6896374E, 410565DD (deleted); 3 change(s) of ours the network did not make; 6 object(s) differed in content	no disposition recorded
R-28	—	#106,782,965	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-29	—	#106,784,160	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 ledger change(s) the network made were absent from ours: FB4F0598	fixed: F181, commit 30d4e7a, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-30	—	#106,786,929	Step 1 · native engine	2 transaction result(s) differed from rippled's; 6 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F182, commit 944aeb1, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-31	—	#106,786,947	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F182, commit 944aeb1, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-32	—	#106,786,948	Step 1 · native engine	2 transaction result(s) differed from rippled's; 6 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-33	—	#106,786,966	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-34	—	#106,786,967	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-35	—	#106,786,968	Step 1 · native engine	1 transaction result(s) differed from rippled's; 3 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-36	—	#106,786,986	Step 1 · native engine	2 transaction result(s) differed from rippled's; 6 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-37	—	#106,788,656	Step 1 · native engine	1 object(s) differed in content	fixed: F183, commit 5449dfa, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-38	—	#106,791,885	Step 1 · native engine	1 ledger change(s) the network made were absent from ours: 353BBCFB; 2 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: F184, commit 199e7d4, deployed in cycle 121 (2026-09-11 17:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-05-01, INC-01 is INC-2026-09-05-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
14:26 UTC	xrpl-validator	`f314dca`	fix(ledger): a TrustSet toward a missing issuer is tecNO_DST (finding 161)
14:32 UTC	xrpl-validator	`039e57e`	fix(ledger): a payment cannot ripple through an issuer whose NoRipple flag sits on both lines (finding 162)
14:53 UTC	xrpl-validator	`ddbc941`	fix(ledger): fence the C++ excerpt in check_no_ripple's docs so rustdoc does not compile it
15:39 UTC	xrpl-validator	`fd8379b`	fix(ledger): a token escrow snapshots the issuer's transfer rate and finishes net of it (finding 164)
15:39 UTC	xrpl-validator	`20bd643`	fix(ledger): a live tip beyond the strict limit but inside the inflated one is attempted, not a stop (finding 163)

Merged	Repo	Commit	Subject
16:27 UTC	xrpl-validator	`dd4c94e`	fix(ledger): an IOU source is funded from its original balance less its debits, credits deferred (finding 165)
17:12 UTC	xrpl-validator	`6283d40`	fix(ledger): a tfSell past its minimum still rests the unsold remainder, judged after repricing (finding 172)
17:12 UTC	xrpl-validator	`c1c1980`	fix(ledger): an issuer-owned maker still grosses the taker's spend by the transfer rate (finding 171)
17:12 UTC	xrpl-validator	`d00a396`	fix(ledger): a book tip beyond the limit does not drop the direct strand when the pool's own bound clears it (finding 169a)
17:12 UTC	xrpl-validator	`128b564`	fix(ledger): an LP-token line is torn down only in rippleCreditIOU's full default state (finding 168)
17:12 UTC	xrpl-validator	`48928e9`	fix(ledger): TrustSet creates lines with auth/freeze/deep-freeze bits and enforces the freeze invariants (finding 167)
17:12 UTC	xrpl-validator	`51a6221`	fix(ledger): the AMM limit judge follows limitOut alone and tip()'s quality function (finding 166)
17:55 UTC	xrpl-validator	`059b37e`	fix(ledger): a payment's sender is bounded by the deferred-debits fold, and a funds-limited fill clamps to the offer's own TakerPays (findings 169b, 169c)
17:55 UTC	xrpl-validator	`fab8f1e`	fix(ledger): a forward surplus keeps the reverse cache, a crossing's remainingOut folds to sixteen digits, and the sender bound is the SendMax hop's alone (findings 170, 173, 169b)
17:55 UTC	xrpl-validator	`188fda9`	fix(ledger): an LP-token withdrawal snaps the requested tokens to the pool balance's precision (finding 175)
17:55 UTC	xrpl-validator	`ceabbf8`	fix(ledger): the bridged leg's remainingOut folds too, and a line-bound fill pays the bound's gross verbatim (findings 173, 174)
18:40 UTC	xrpl-validator	`8a23425`	fix(ledger): a carry hop takes back no more than the chain parked on the sender's line (finding 177)
18:40 UTC	xrpl-validator	`5f4bf34`	fix(ledger): undoing a park-created line also reverts its creator's OwnerCount (finding 178)
18:40 UTC	xrpl-validator	`008363c`	fix(ledger): a bridge leg's offer ladder runs to rippled's 1000-offer limit (finding 179)
18:40 UTC	xrpl-validator	`dda33ab`	fix(ledger): a pool slice anchored at a level's own quality shares that level's pass (finding 180)
19:33 UTC	xrpl-validator	`962e81b`	fix(scripts): fetch_tx_bundle seeds explicit-path books and pools, 200-offer book depth
19:33 UTC	xrpl-validator	`30d4e7a`	fix(xrpl-ledger): F181 — a holder-less sender ISSUES to the issuer within its limit
21:53 UTC	xrpl-validator	`944aeb1`	fix(xrpl-ledger): F182 — CredentialCreate needs the reserve for one more object
23:48 UTC	xrpl-validator	`5449dfa`	fix(xrpl-ledger): F183 — a strand's spend is the walk's own figure, not the line's quantised delta
03:14 UTC	xrpl-validator	`199e7d4`	fix(xrpl-ledger): F184 — an autobridged leg crosses the taker's own offer

Findings fixed

- **fix(xrpl-ledger): F181 — a holder-less sender ISSUES to the issuer within its limit** (`30d4e7a`, xrpl-validator) `apply_iou_direct` sized the sender's leg by `available()` — the sender's HOLDINGS of the issuer's IOU — so a sender holding none was tecPATH_DRY. rippled's sender hop is `DirectStepI::maxPaymentFlow` (DirectStep.cpp:476-490): when `srcOwed <= 0` the hop ISSUES the sender's own IOU up to `creditLimit2(dst, src) + srcOwed`, the issuer's trust limit toward the sender less what the sender already owes it.

- **fix(xrpl-ledger): F182 — CredentialCreate needs the reserve for one more object** (`944aeb1`, xrpl-validator) The native CredentialCreate had no reserve check and no subject check: it wrote the credential, both owner-directory pages and the OwnerCount bump for any issuer. rippled (CredentialCreate.cpp) answers tecNO_TARGET when the subject account does not exist (preclaim, before the duplicate test) and, in doApply before any write, `checkReserve(ctx, sleIssuer, preFeeBalance_, {.ownerCountDelta = 1})` — the balance BEFORE …
- **fix(xrpl-ledger): F183 — a strand's spend is the walk's own figure, not the line's quantised delta** (`5449dfa`, xrpl-validator) `strand_pass` measured what a strand spent by differencing the sender's line before and after, trusting the walk's own figure only when the two agreed to 1e-9 relative. A trust line holds 16 significant digits on its MAGNITUDE, so the delta is the true spend quantised to the balance's ulp — for a small spend from a large line that quantum dwarfs the relative test and the exact figure was refused. rippled never reads …
- **fix(xrpl-ledger): F184 — an autobridged leg crosses the taker's own offer** (`199e7d4`, xrpl-validator) The self-cross rule (`BookOfferCrossingStep::limitSelfCrossQuality`, BookStep.cpp:402-445) removes the taker's own offer only on the DEFAULT path — `defaultPath_ && offer.quality() >= qualityThreshold_ && owner == strandSrc == strandDst` — so on an autobridged leg the taker's own offer is crossed like any other maker's, its self-transfers netting to nothing (`accountSend` with sender == receiver is a no-op).

Other engineering changes

- **fix(ledger): a TrustSet toward a missing issuer is tecNO_DST (finding 161)** (`f314dca`, xrpl-validator) Ledger #106759500, tx D92BD1028719, and six more from the same account that day: rMY7ERFxNN sets a 100000 "RLUSD" line toward rQhWct2f..., an account that does not exist. rippled's TrustSet::preclaim reads the destination root and, with AMM or SingleAssetVault enabled, returns tecNO_DST when it is missing (TrustSet.cpp:213-216); both amendments are live on mainnet, so mainnet takes the fee and writes nothing else.
- **fix(ledger): a payment cannot ripple through an issuer whose NoRipple flag sits on both lines (finding 162)** (`039e57e`, xrpl-validator) Ledger #106779252, tx 3AD7EA863312, and six more that morning: rJEvC4cuk pays 2,000,000 XRPFLORIDAGATORS to rpFYjv6SF. Both trust lines exist, the sender holds 99,999,999,999 and the destination's limit is ample, yet mainnet returns tecPATH_DRY and charges only the fee. The issuer, rMpjR1oh, carries its NoRipple flag on its side of both holders' lines, and the default path ripples through the issuer: …
- **fix(ledger): fence the C++ excerpt in check_no_ripple's docs so rustdoc does not compile it** (`ddbc941`, xrpl-validator) gate67 failed one test of 659: rustdoc took the indented rippled excerpt in finding 162's doc comment for a Rust doctest. A text fence keeps it prose.
- **fix(ledger): a token escrow snapshots the issuer's transfer rate and finishes net of it (finding 164)** (`fd8379b`, xrpl-validator) Ledger #106758324, tx 330B52F56E04: r4uNAZYC2k escrows 10 BST under a crypto-condition; BST's issuer charges a 1.08 transfer rate. Ledger #106758384, tx F578BE463712: rpcjmdWFCQ finishes it with the fulfillment.
- **fix(ledger): a live tip beyond the strict limit but inside the inflated one is attempted, not a stop (finding 163)** (`20bd643`, xrpl-validator) Ledger #106771950, tx B1283A3C1288, again at #106772087 and #106772093: rBTwLga3i2 buys 1,000,000 XRP with 1,380,000.5 USD.Bitstamp, a limit of 1.3800005e-6 USD per drop. The book's tip is rsRdjxq2's 2208-drop dust offer, FILED at 1.3800013e-6 (a hair beyond the strict limit, inside the 1.0015-inflated one) while its own amounts, 0.0030459516 USD for 2208 drops, price at 1.3795e-6, inside the strict limit. rippled …

- **fix(ledger): an IOU source is funded from its original balance less its debits, credits deferred (finding 165)** (`dd4c94e`, xrpl-validator) Ledger #106772946, tx C9E92CF8532F: rLc1HmTpWg's circular partial payment (it pays itself) buys FUZZY through a book that holds its own offers, so one trust line is credited as the destination and debited as the maker inside a single transaction.
- **fix(ledger): a tfSell past its minimum still rests the unsold remainder, judged after repricing (finding 172)** (`6283d40`, xrpl-validator) #106758110 E664290A: 7559 BAYN tfSell; the pool took 7456.036 for 124.14 XRP — past the 118.68 minimum — and mainnet rested 102.963884945252 BAYN for 1616532 drops. rippled judges "Offer fully crossed" on the amounts that would rest (OfferCreate.cpp:778), and for tfSell the out side is re-derived from the unsold in first (flowCross :489-506, afterCross.out = divRoundStrict(afterCross.in, rate)). Our guard read the …
- **fix(ledger): an issuer-owned maker still grosses the taker's spend by the transfer rate (finding 171)** (`c1c1980`, xrpl-validator) #106758320 A092BA66 (tfSell|tfIoC selling 18.5185185185186 BST at rate 1.08 into the ISSUER's own offer): the in-side rate is waived only when the offer's owner is the strand source — alice paying alice (BookOfferCrossingStep::getOfrInRate) — never because the owner is the issuer. The taker's DirectStep hands the issuer the gross; the issuer→owner send is the no-op. Mainnet debits the taker 20.0000 and the issuer's …
- **fix(ledger): a book tip beyond the limit does not drop the direct strand when the pool's own bound clears it (finding 169a)** (`d00a396`, xrpl-validator) #106766924 961C38C7 (IoC XLM→CNY, iteration 1): activateNext's bound is tip() — the AMM offer when it beats the LOB tip, else the LOB tip (BookStep.cpp:921-953). A LOB tip beyond the limit makes qualityThreshold nullopt, so getOffer emits the maxOffer (quality = the feeless spot) or the Fibonacci slice, and the strand lives while that bound clears the limit. The next direct offer sat at 0.83737 XLM/CNY against the …
- **fix(ledger): an LP-token line is torn down only in rippleCreditIOU's full default state (finding 168)** (`128b564`, xrpl-validator) #106766906 29526F1B (AMMWithdraw tfLPToken): rEWLR7's LP line carried lsfLowNoRipple while the account has DefaultRipple set — a non-default line — so mainnet zeroed and KEPT it (OwnerCount 709 untouched). We deleted it and took 708. tear_down_lp_line checked only the limit and the qualities; it now applies the whole guard (TokenHelpers.cpp:702-716): the LP's reserve bit set, its NoRipple bit differing from the …
- **fix(ledger): TrustSet creates lines with auth/freeze/deep-freeze bits and enforces the freeze invariants (finding 167)** (`48928e9`, xrpl-validator) #106761226 63F07E36 and #106773044 05F705D5: an issuer authorising a holder that has no line yet CREATES it with lsfLowAuth — trustCreate carries bAuth, bFreeze and bDeepFreeze onto the creator's side (RippleStateHelpers.cpp:262-277, TrustSet.cpp:735-738). Our creation path only minted the reserve and NoRipple bits: mainnet 0x250000, ours 0x210000, invisible to every per-tx leg but the replay's byte diff.
- **fix(ledger): the AMM limit judge follows limitOut alone and tip()'s quality function (finding 166)** (`51a6221`, xrpl-validator) Two crossings the deploy66 soak caught in the single-path AMM tail: a) #106759265 7556BE4A (IoC sell): adjustedRemOut is settled by limitOut ALONE (StrandFlow.h:664-672) before the passes cap the input. We reset `adjusted` on the input clamp, judged the clamped fill strictly and killed a 1.4e-14 miss rippled forgives (tecKILLED vs tesSUCCESS).
- **fix(ledger): a payment's sender is bounded by the deferred-debits fold, and a funds-limited fill clamps to the offer's own TakerPays (findings 169b, 169c)** (`059b37e`, xrpl-validator) 169b — #106777783 6EF9AF7D and #106773978 D33DB4C8 (tfPartialPayment IOU→XRP sweeps): rippled's per-iteration DirectStep bound is PaymentSandbox::balanceHookIOU, min(live, origBalance − Σdebits) with the debits a 16-digit half-even fold.

Its trace shows the last DirectStepI::rev bounded at 50251.6343549063 while the line reads 50251.63435490632, and the line keeps 2e-11. The walk's line-bound (held less this level's …

- **fix(ledger): a forward surplus keeps the reverse cache, a crossing's remainingOut folds to sixteen digits, and the sender bound is the SendMax hop's alone (findings 170, 173, 169b)** (`fab8f1e` , xrpl-validator) 170 — #106763015 360F1D30 (1e6 drops → GBP through the pool, issuer rate 1.002): DirectStepI::setCacheLimiting keeps the REVERSE cache whenever the forward pass would exceed it — fwdIn > cache.in, srcToDst or out likewise. rippled's trace: rev asks 1.678428412228995 for 1.67507825571756, the pool's forward swap yields 1.678428412229, and the credit is the cache's 1.67507825571756 (fwd srcToDst 1.675078255717565 …
- **fix(ledger): an LP-token withdrawal snaps the requested tokens to the pool balance's precision (finding 175)** (`188fda9` , xrpl-validator) #106782285 A4B18D75 (AMMWithdraw tfLPToken): fixAMMv1_3's adjustLPTokens (AMMHelpers.cpp:181-189, via AMMWithdraw.cpp:752-761, WithdrawAll exempt) rounds the request to what the pool's LPTokenBalance can represent — `(lpTokens − lptAMMBalance) + lptAMMBalance` under Downward rounding. rPLvYSKR asked 38534539.5663724 of a 1143652631.196906 pool; rippled burns 38534539.566372, the LP keeps 0.0000004 and its line …
- **fix(ledger): the bridged leg's remainingOut folds too, and a line-bound fill pays the bound's gross verbatim (findings 173, 174)** (`ceabbf8` , xrpl-validator) 173, bridged leg — #106781871 825A156A (XAH→RLUSD crossing through the XRP bridge): the crossing's remainingOut is a sixteen-digit IOUAmount on the bridged leg as on the direct walk. 1.84386153980520|36 folds to …5204, and rippled's last maker gives 0.528031539805204; the exact remainder had us take …52036 and left the offer 4e-16 apart.
- **fix(ledger): a carry hop takes back no more than the chain parked on the sender's line (finding 177)** (`8a23425` , xrpl-validator) #106733664 37F54060 and #106735554 9BCDD090 (rapido, mixed paths): a carry hop spends what the hop chain parked on the sender's line (finding 98) — a line rippled's strand never touches, the carry riding the intermediaries' lines instead (q127's trace: rpxqUyfn in 0.8379734275750757, fee at rvYAfWj5's issuer→maker send, 3.3e-15 left on rpxqUyfn). Re-grossing the net take-back landed one ulp over the park and …
- **fix(ledger): undoing a park-created line also reverts its creator's OwnerCount (finding 178)** (`5f4bf34` , xrpl-validator) #106735554 9BCDD090 (rapido, a mixed path whose first book feeds an account run): the mixed driver measures the carry by parking it on the sender's line and undoes the park before the run. A line the park CREATED carries its creator's reserve bit, sits in both owner directories and charged the creator's OwnerCount (line_adjust's trustCreate path); undo_inflight_lines forgot the line and its directory entries but …
- **fix(ledger): a bridge leg's offer ladder runs to rippled's 1000-offer limit (finding 179)** (`008363c` , xrpl-validator) kMaxOffersToConsume is 1000 offers per BookStep per transaction (BookStep.cpp:60, :1085/:1247); cross_bridged built each leg's ladder from the first 128. Surfaced by #106743104 F8084760, a 54-iteration XRPH→XRP→RLUSD crawl whose bundle probe went dry at round 50 — that probe turned out to lack the XRPH/XRP book tip rippled anchors the pool to (the offer is never consumed, so it never reaches the metadata or the …
- **fix(ledger): a pool slice anchored at a level's own quality shares that level's pass (finding 180)** (`dda33ab` , xrpl-validator) #106770629 FF3E66E6 (XRP→USDC→RLUSD partial payment): rippled's forEachOffer runs tryAMM on the funded tip's quality and then every offer AT that quality in the same pass — ofrQ is the anchored AMM offer's quality, the tip's — so the pool's 28.0574113498 USDC and rLtCVnoj's 1.538480007399999 reach the RLUSD book as one 29.5958913572 and its maker is debited once: 7879.338142785851 − 29.59331668965259 = …198. The …
- **fix(scripts): fetch_tx_bundle seeds explicit-path books and pools, 200-offer book depth** (`962e81b` , xrpl-validator) Two bundle blind spots that read as engine residues in the s/q sweeps: * EXPLICIT PATHS. Only the

direct and XRP-bridge books were seeded, so a Payment whose `Paths` hop through other currencies had no book at all in the bundle and the probe ran the strand DRY. #106758148 0638A180 (XRP → PKEG → FARM) and #106766924 DA053F84 (USDC → PLX → XLM → XRP) — s332/s367 — are green once every currency hop's book (an account hop …

Independence program

No independence record for UTC day 2026-09-05; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

Open items

- Receipt R-02 (#106,771,926): open, under investigation.
- Receipt R-04 (#106,771,970): open, under investigation.
- Receipt R-11 (#106,774,868): open, under investigation.
- Receipt R-13 (#106,778,558): open, under investigation.
- Receipt R-15 (#106,779,295): open, under investigation.
- Receipt R-16 (#106,779,421): open, under investigation.
- Receipt R-17 (#106,779,797): open, under investigation.
- Receipt R-18 (#106,779,825): open, under investigation.
- Receipt R-19 (#106,779,836): open, under investigation.
- Receipt R-20 (#106,779,890): open, under investigation.
- Receipt R-21 (#106,780,372): open, under investigation.
- Receipt R-22 (#106,780,685): open, under investigation.
- Receipt R-23 (#106,780,690): open, under investigation.

- Receipt R-27 (#106,782,599): open, under investigation.
- Receipt R-28 (#106,782,965): open, under investigation.
- Receipt R-32 (#106,786,948): open, under investigation.
- Receipt R-33 (#106,786,966): open, under investigation.
- Receipt R-34 (#106,786,967): open, under investigation.
- Receipt R-35 (#106,786,968): open, under investigation.
- Receipt R-36 (#106,786,986): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Deferred Ledger-Triggered Transaction Execution** (payments) Add a new owner-reserved ledger object ltDEFERRED_TX (keylet::deferredTx(account, sequence)) created by a new DeferredTxCreate transaction, storing a
- **Live Credential Re-Check on MPT Transfer** (compliance) Add an lsfRequireTransferCredential flag and sfTransferCredentialIDs SField (STVector256, max 8 keylets) to MPTokenIssuance, settable only by the issu
- **Owner Directory Tail Compaction Sweep** (fees) Add a new transaction ttDIRECTORY_COMPACT that walks up to 32 entries from the tail (oldest) pages of the invoker's own OwnerDirectory and deletes any

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-05 14:26	09-05 14:26	xrpl-validator	`f314dca`	fix(ledger): a TrustSet toward a missing issuer is tecNO_DST (finding 161)
09-05 14:32	09-05 14:32	xrpl-validator	`039e57e`	fix(ledger): a payment cannot ripple through an issuer whose NoRipple flag sits on both lines (finding 162)
09-05 14:53	09-05 14:53	xrpl-validator	`ddbc941`	fix(ledger): fence the C++ excerpt in check_no_ripple's docs so rustdoc does not compile it
09-05 15:39	09-05 15:39	xrpl-validator	`fd8379b`	fix(ledger): a token escrow snapshots the issuer's transfer rate and finishes net of it (finding 164)
09-05 15:39	09-05 15:39	xrpl-validator	`20bd643`	fix(ledger): a live tip beyond the strict limit but inside the inflated one is attempted, not a stop (finding 163)
09-05 16:27	09-05 16:27	xrpl-validator	`dd4c94e`	fix(ledger): an IOU source is funded from its original balance less its debits, credits deferred (finding 165)
09-05 17:12	09-05 17:12	xrpl-validator	`6283d40`	fix(ledger): a tfSell past its minimum still rests the unsold remainder, judged after repricing (finding 172)

Committed	Merged into main	Repo	Commit	Subject
09-05 17:12	09-05 17:12	xrpl-validator	`c1c1980`	fix(ledger): an issuer-owned maker still grosses the taker's spend by the transfer rate (finding 171)
09-05 17:12	09-05 17:12	xrpl-validator	`d00a396`	fix(ledger): a book tip beyond the limit does not drop the direct strand when the pool's own bound clears it (finding 169a)
09-05 17:12	09-05 17:12	xrpl-validator	`128b564`	fix(ledger): an LP-token line is torn down only in rippleCreditIOU's full default state (finding 168)
09-05 17:12	09-05 17:12	xrpl-validator	`48928e9`	fix(ledger): TrustSet creates lines with auth/freeze/deep-freeze bits and enforces the freeze invariants (finding 167)
09-05 17:12	09-05 17:12	xrpl-validator	`51a6221`	fix(ledger): the AMM limit judge follows limitOut alone and tip()'s quality function (finding 166)
09-05 17:15	09-05 17:55	xrpl-validator	`059b37e`	fix(ledger): a payment's sender is bounded by the deferred-debits fold, and a funds-limited fill clamps to the offer's own TakerPays (findings 169b, 169c)
09-05 17:18	09-05 17:55	xrpl-validator	`fab8f1e`	fix(ledger): a forward surplus keeps the reverse cache, a crossing's remainingOut folds to sixteen digits, and the sender bound is the SendMax hop's alone (findings 170, 173, 169b)
09-05 17:39	09-05 17:55	xrpl-validator	`188fda9`	fix(ledger): an LP-token withdrawal snaps the requested tokens to the pool balance's precision (finding 175)
09-05 17:39	09-05 17:55	xrpl-validator	`ceabbf8`	fix(ledger): the bridged leg's remainingOut folds too, and a line-bound fill pays the bound's gross verbatim (findings 173, 174)
09-05 18:04	09-05 18:40	xrpl-validator	`8a23425`	fix(ledger): a carry hop takes back no more than the chain parked on the sender's line (finding 177)
09-05 18:13	09-05 18:40	xrpl-validator	`5f4bf34`	fix(ledger): undoing a park-created line also reverts its creator's OwnerCount (finding 178)
09-05 18:18	09-05 18:40	xrpl-validator	`008363c`	fix(ledger): a bridge leg's offer ladder runs to rippled's 1000-offer limit (finding 179)
09-05 18:25	09-05 18:40	xrpl-validator	`dda33ab`	fix(ledger): a pool slice anchored at a level's own quality shares that level's pass (finding 180)
09-05 19:32	09-05 19:33	xrpl-validator	`962e81b`	fix(scripts): fetch_tx_bundle seeds explicit-path books and pools, 200-offer book depth
09-05 19:32	09-05 19:33	xrpl-validator	`30d4e7a`	fix(xrpl-ledger): F181 — a holder-less sender ISSUES to the issuer within its limit
09-05 21:53	09-05 21:53	xrpl-validator	`944aeb1`	fix(xrpl-ledger): F182 — CredentialCreate needs the reserve for one more object
09-05 23:48	09-05 23:48	xrpl-validator	`5449dfa`	fix(xrpl-ledger): F183 — a strand's spend is the walk's own figure, not the line's quantised delta
09-06 03:14	09-06 03:14	xrpl-validator	`199e7d4`	fix(xrpl-ledger): F184 — an autobridged leg crosses the taker's own offer

Times in UTC. A commit appears on the day it was committed to main or merged into it.